

KUANDIAN, China

WMO: 544930

Lat: 40.7088N

Lon: 124.7817E

Elev: 261

StdP: 98.23

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.8	-19.7	-27.3	0.3	-17.5	-25.0	0.4	-16.1	6.1	-9.3	5.4	-8.7	0.7	200	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	30.5	22.6	29.0	22.0	27.7	21.5	24.9	27.9	24.1	26.6	23.4	25.7	2.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.6	26.5	23.4	18.8	25.6	22.7	18.0	24.8	77.6	28.0	74.4	26.6	71.6	25.7	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.9	5.1	4.4	-24.6	32.9	3.0	1.4	-26.7	33.9	-28.4	34.7	-30.1	35.5	-32.3	36.5		
			-24.8	25.8	2.9	1.2	-26.9	26.7	-28.6	27.4	-30.3	28.1	-32.4	28.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-10.2	-6.1	0.8	8.6	15.0	20.0	23.3	22.9	17.3	9.7	0.6	-7.7
	DBStd	12.23	4.54	4.77	4.03	3.91	2.94	2.35	1.95	2.35	3.05	3.93	5.23	4.76
	HDD10.0	2322	627	451	284	71	2	0	0	0	1	53	283	550
	HDD18.3	4187	885	684	542	292	109	11	0	2	54	269	532	808
	CDD10.0	1560	0	0	1	29	156	299	412	399	219	43	2	0
	CDD18.3	383	0	0	0	0	5	60	153	142	23	0	0	0
	CDH23.3	2444	0	0	0	3	94	429	871	897	146	3	0	0
	CDH26.7	534	0	0	0	0	9	94	218	199	13	0	0	0
Wind	WSAvg	1.6	1.3	1.6	1.9	2.1	1.9	1.5	1.4	1.4	1.3	1.4	1.5	1.3
Precipitation	PrecAvg	1099	11	14	19	54	73	107	314	302	95	52	37	15
	PrecMax	1815	50	37	48	124	167	211	883	625	221	116	84	39
	PrecMin	660	0	0	2	9	17	30	47	22	34	0	5	0
	PrecStd	259	10	10	13	32	39	48	170	157	42	27	20	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	8.8	15.3	23.7	28.1	31.7	32.9	32.1	28.2	23.1	15.0	5.2
		MCWB	0.0	3.2	6.9	12.8	16.4	20.4	24.6	24.0	20.7	15.7	10.5	2.7
	2%	DB	1.0	5.4	12.0	21.1	25.9	29.2	30.7	30.1	26.4	20.4	12.7	3.1
		MCWB	-1.4	1.2	5.5	11.6	15.4	19.3	23.4	23.1	19.4	13.7	9.4	1.1
	5%	DB	-0.7	3.4	9.8	18.9	24.1	27.4	29.0	28.7	25.0	18.5	10.7	1.5
		MCWB	-2.8	0.0	3.9	10.1	14.9	19.1	22.8	22.5	18.0	12.4	7.8	-0.3
	10%	DB	-2.4	1.6	7.8	16.7	22.5	25.9	27.6	27.6	23.7	16.6	8.8	-0.1
		MCWB	-4.5	-1.0	3.0	9.1	14.0	18.7	22.5	22.2	17.4	11.6	6.2	-1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.1	4.2	8.5	15.4	19.1	22.3	26.5	25.9	22.6	17.0	12.1	3.8
		MCDB	2.4	7.5	12.4	19.4	23.8	26.8	30.8	29.2	25.6	20.9	13.9	4.7
	2%	WB	-0.9	1.9	6.4	13.0	17.4	21.4	25.1	25.0	21.3	15.4	10.1	1.5
		MCDB	0.7	4.7	10.4	18.8	22.8	25.7	28.5	27.9	24.3	18.7	12.1	2.8
	5%	WB	-2.6	0.5	4.9	11.5	16.2	20.6	24.2	24.3	20.1	14.1	8.4	-0.1
		MCDB	-0.8	2.8	8.4	16.9	21.4	24.8	26.8	26.8	23.2	16.8	10.2	1.2
	10%	WB	-4.4	-0.7	3.5	10.1	15.3	19.9	23.5	23.7	19.0	12.8	6.5	-1.7
		MCDB	-2.6	1.6	7.2	15.1	20.3	24.0	25.8	25.9	21.7	15.4	8.4	-0.2
Mean Daily Temperature Range	5% DB	MDBR	11.8	11.3	10.4	12.1	11.9	9.5	7.1	8.3	11.5	12.0	9.7	10.4
		MCDBR	11.9	12.1	13.7	16.0	15.8	13.8	10.4	10.3	12.6	14.3	11.3	10.5
		MCWBR	9.4	8.1	7.7	7.7	6.7	5.3	4.0	4.3	5.9	7.7	7.6	8.3
	5% WB	MCDBR	10.7	10.2	12.0	13.0	12.4	9.8	7.1	7.0	9.5	11.2	9.2	9.3
MCWBR		8.9	7.4	7.7	7.0	6.0	4.4	3.3	3.4	5.0	7.0	7.0	8.0	
Clear-Sky Solar Irradiance	taub	0.374	0.450	0.513	0.598	0.630	0.701	0.668	0.574	0.504	0.529	0.456	0.386	
	taud	1.984	1.828	1.757	1.649	1.625	1.558	1.673	1.869	1.970	1.807	1.902	1.988	
	Ebn at Noon	731	721	725	696	687	640	657	709	730	636	628	676	
	Edh at Noon	124	170	204	243	254	272	241	193	163	172	135	115	
All-Sky Solar Radiation	RadAvg	2.44	3.20	4.27	4.86	5.49	4.90	4.16	4.42	4.34	3.25	2.24	2.02	
	RadStd	0.13	0.21	0.28	0.35	0.43	0.56	0.44	0.51	0.36	0.20	0.26	0.16	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days			
			99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47)	Regional (13 neighbors)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
		N/A	N/A	-0.94	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page